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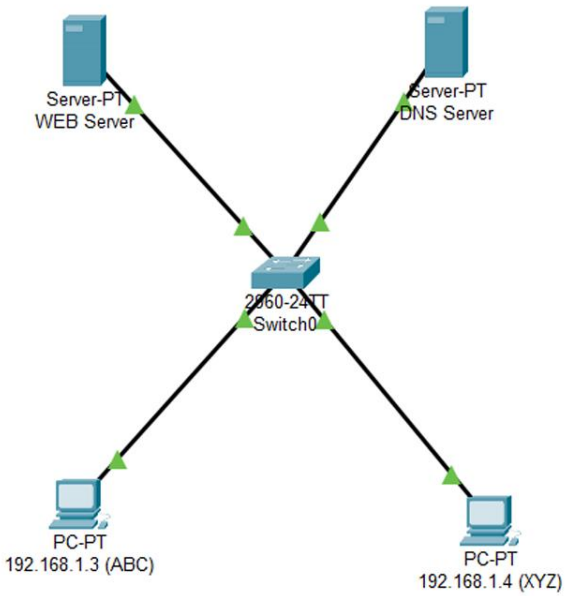
Lab Practical #06:

Study the application layer protocol DNS, DHCP, FTP.

Practical Assignment #06:

1. **Implement the application layer protocol DNS, DHCP, and FTP. Also check connectivity between them using ping command or PDU utility.**

➔ **DNS Sever**

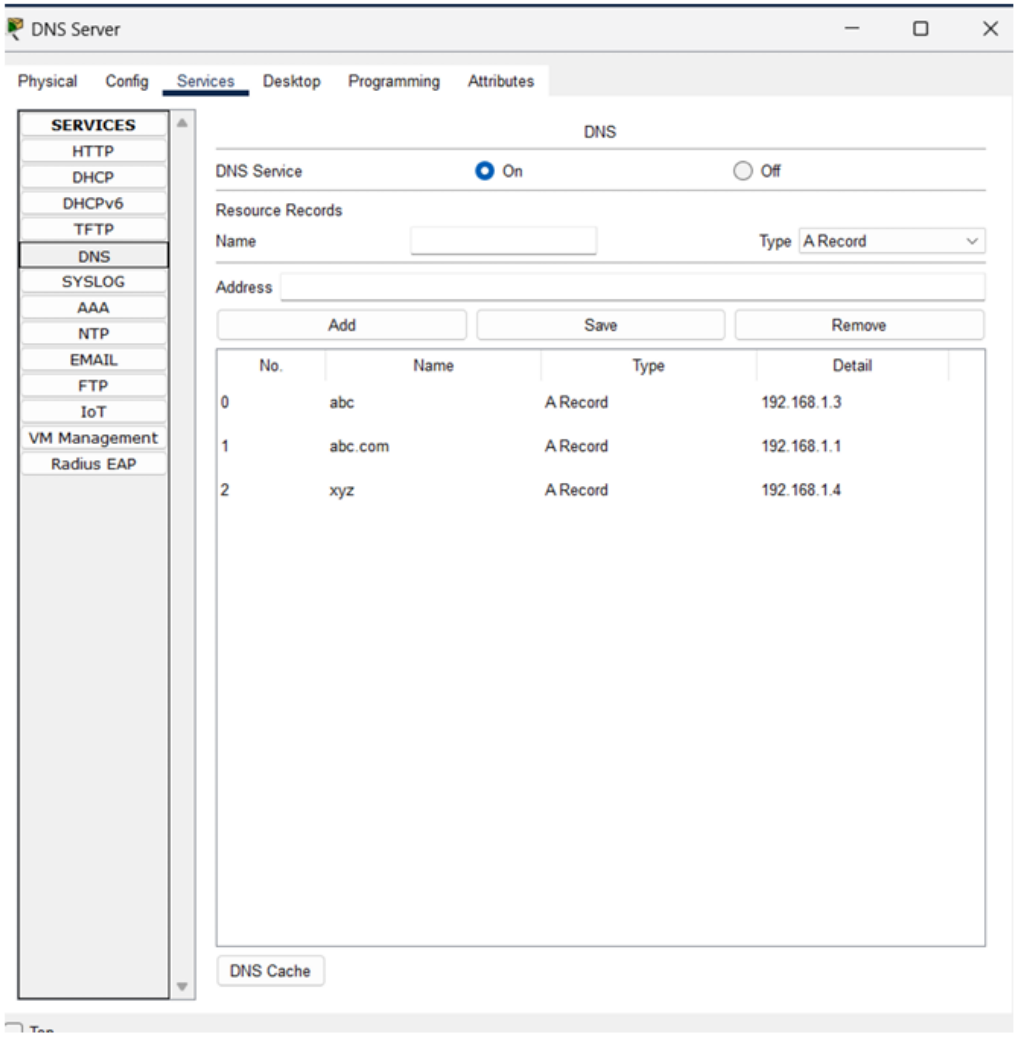


1.1 Circuit Design of DNS

IP Address:

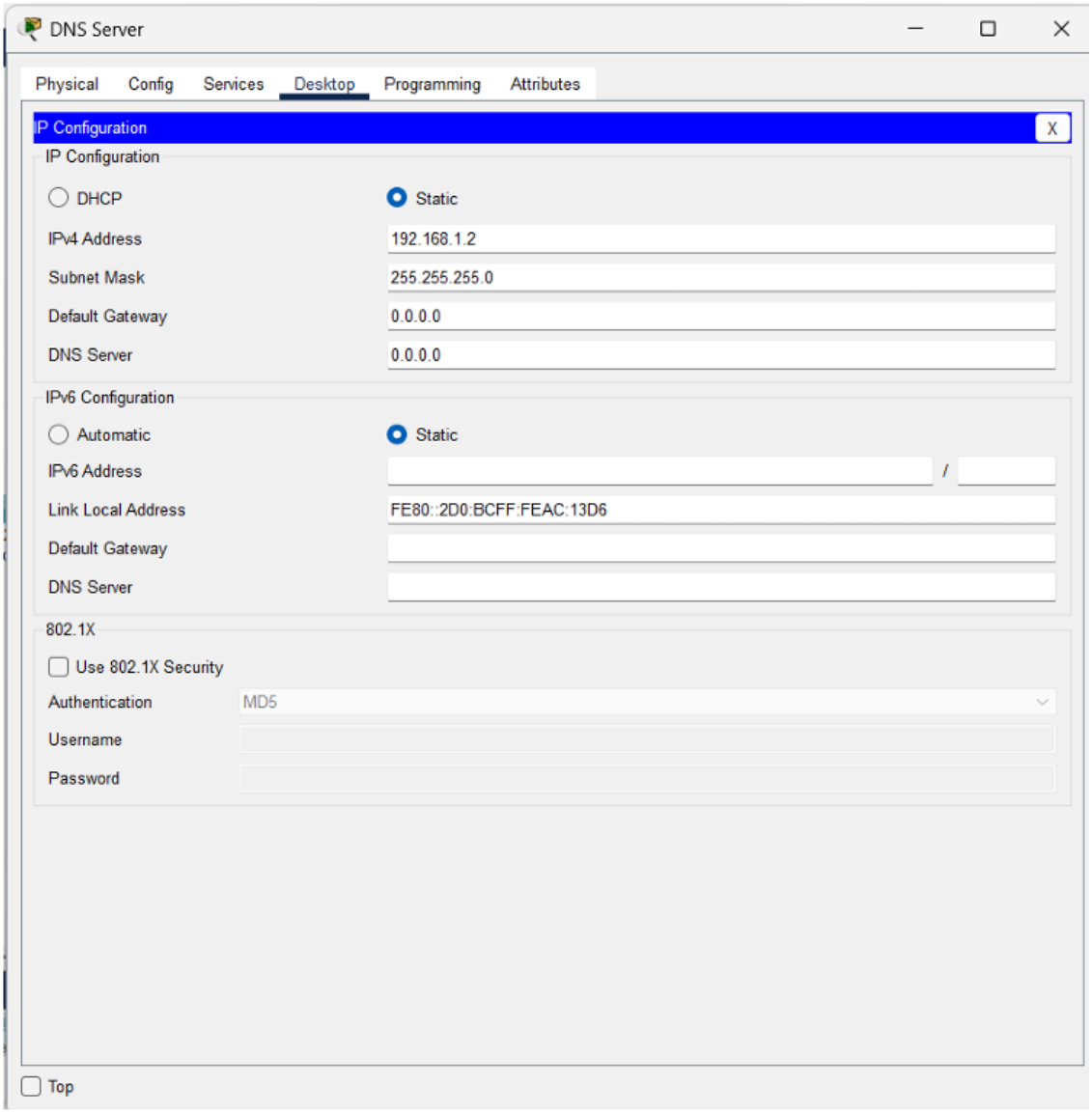
Device	IP Address	Subnet Mask	Default Gateway	DNS Server
WEB Server	192.168.1.2	255.255.255.0	192.168.1.1	192.168.1.5
DNS Server	192.168.1.5	255.255.255.0	192.168.1.1	192.168.1.5
PC0 (ABC)	192.168.1.3	255.255.255.0	192.168.1.1	192.168.1.5
PC1 (XYZ)	192.168.1.4	255.255.255.0	192.168.1.1	192.168.1.5

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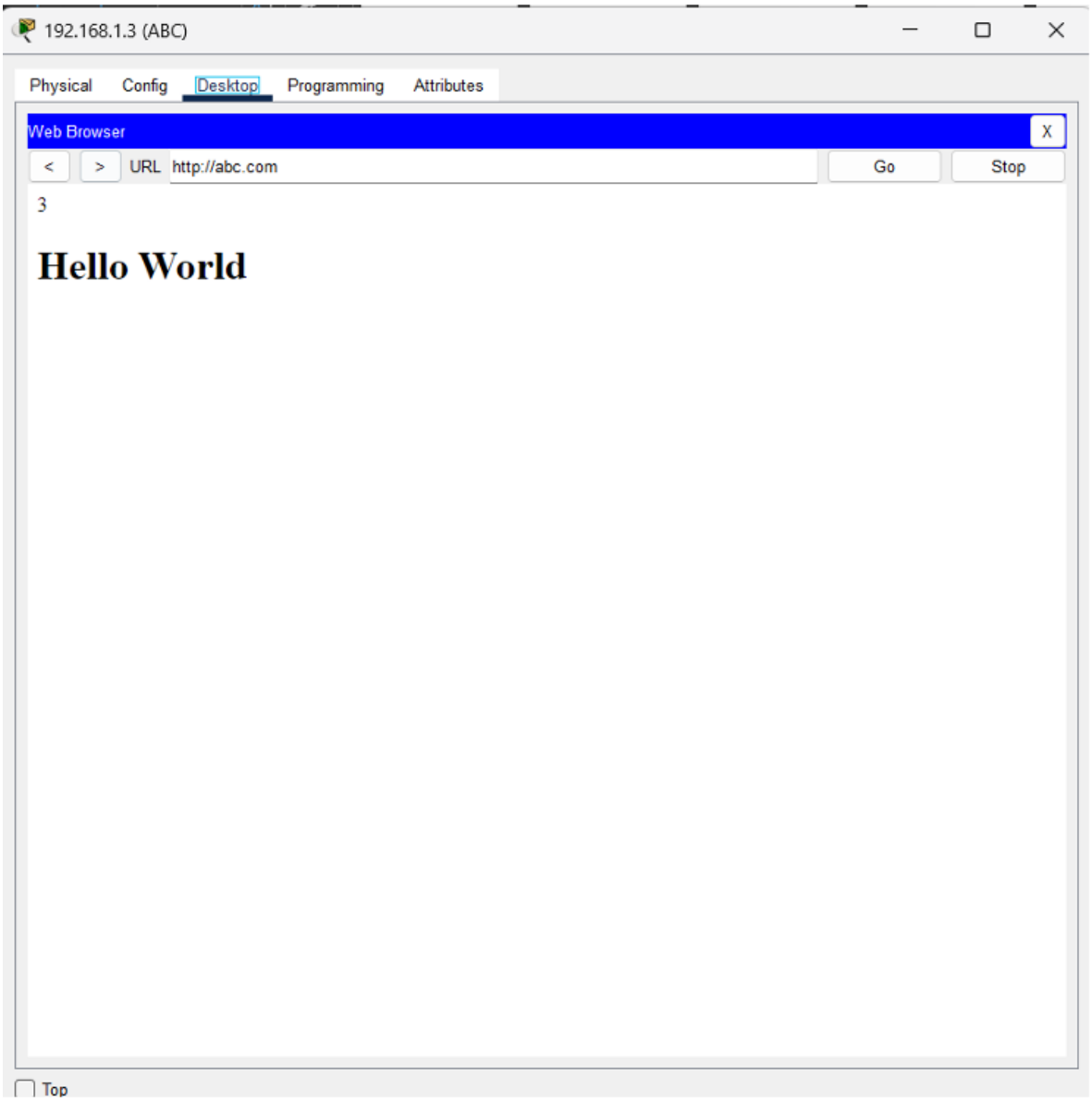
1.2 DNS Server Service

Date: 16/07/2026



1.3 DNS Server IP Configuration

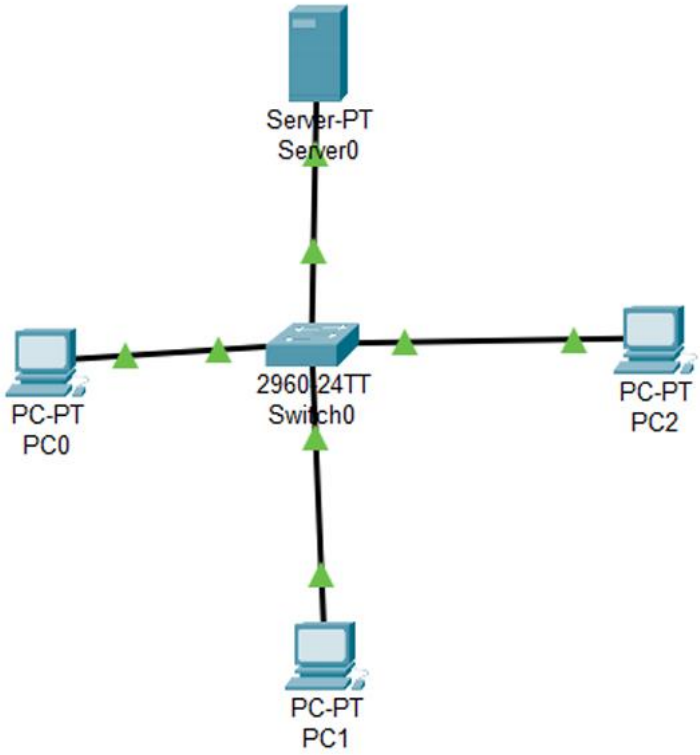
Date: 16/07/2026



1.4 Domain Name through access site

Date: 16/07/2026

➔ **DHCP Sever:**

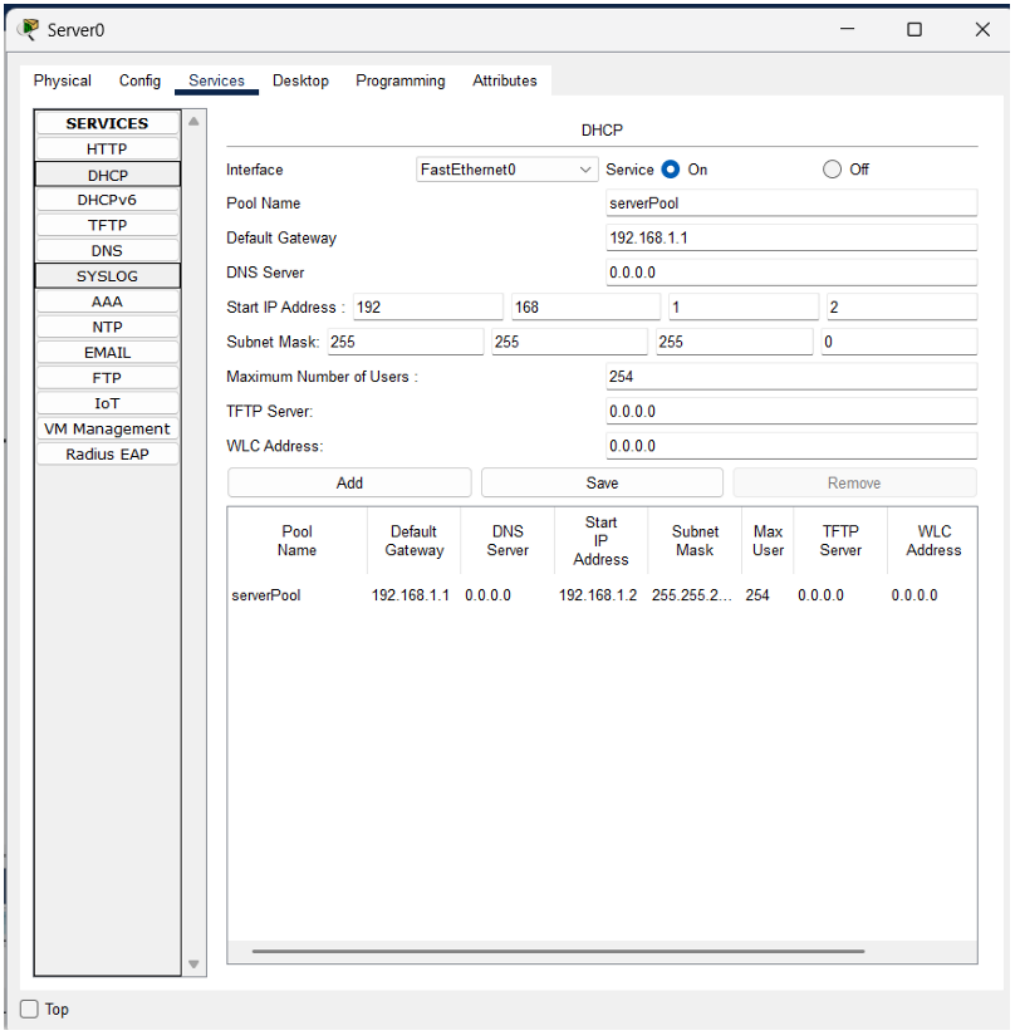


2.1 Circuit Design of DHCP

Date: 16/07/2026

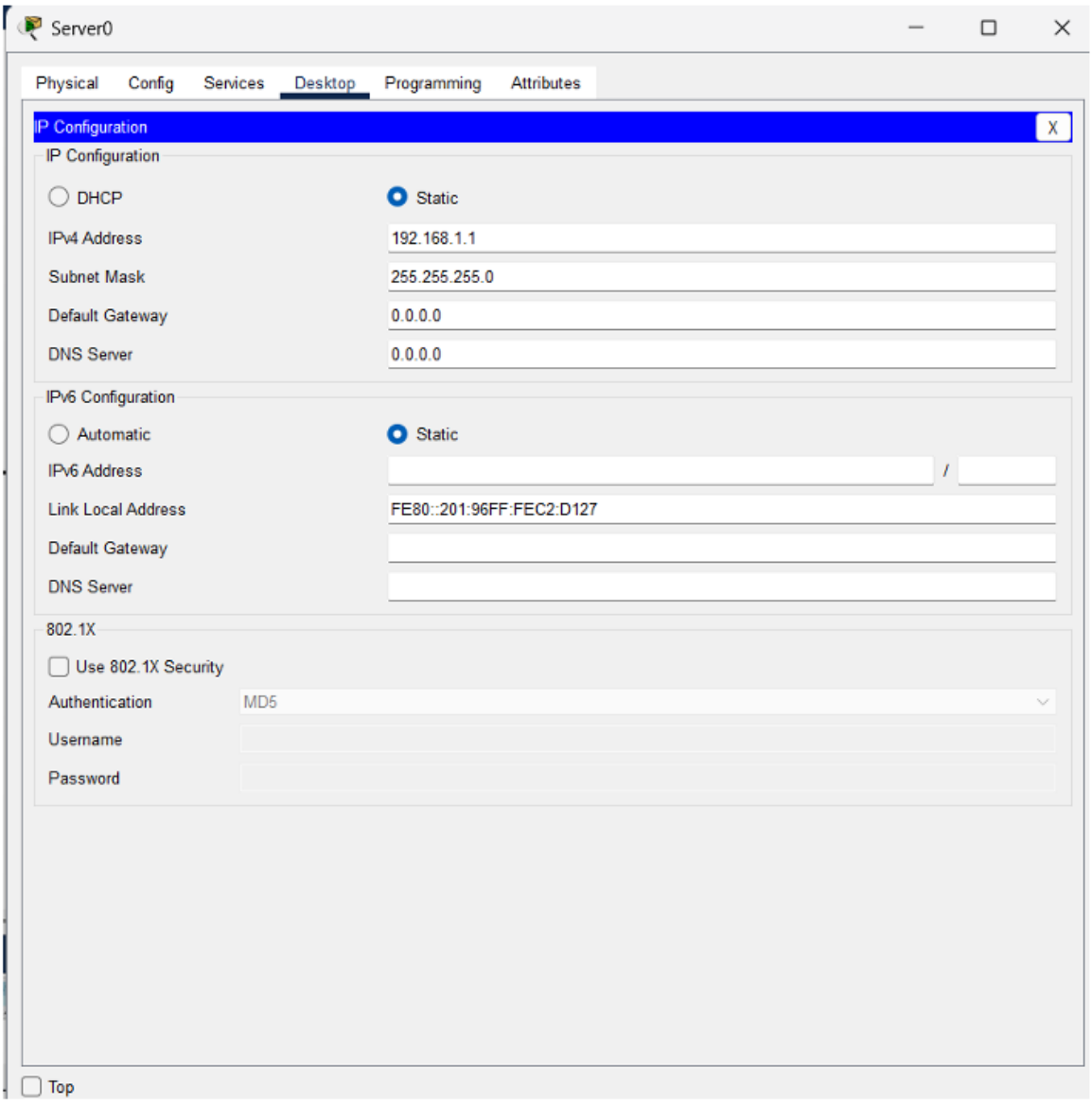
IP Address Table:

Device	IP Address	Subnet Mask	Default Gateway
Server0	192.168.10.2	255.255.255.0	192.168.10.1
PC0	192.168.10.10 *(DHCP)*	255.255.255.0	192.168.10.1
PC1	192.168.10.11 *(DHCP)*	255.255.255.0	192.168.10.1
PC2	192.168.10.12 *(DHCP)*	255.255.255.0	192.168.10.1



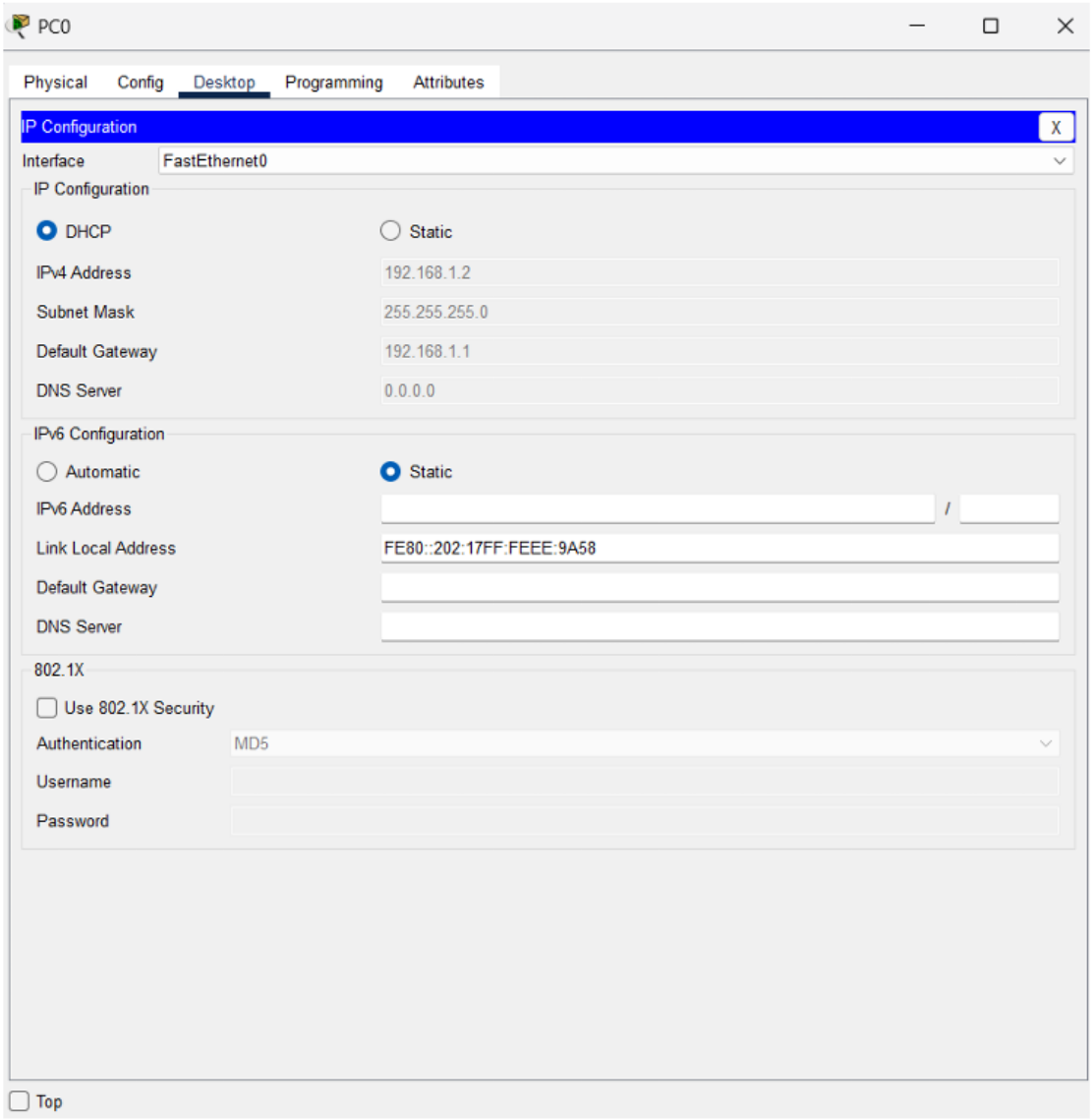
2.2 DHCP Server Service

Date: 16/07/2026



2.3 DHCP Server IP Configuration

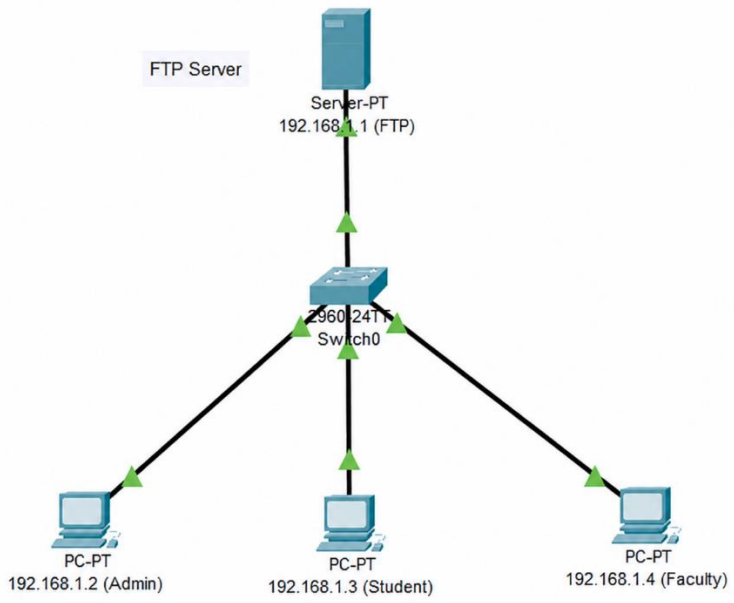
Date: 16/07/2026



2.4 DHCP Devices IP Configuration

Date: 16/07/2026

➔ **FTP Server:**

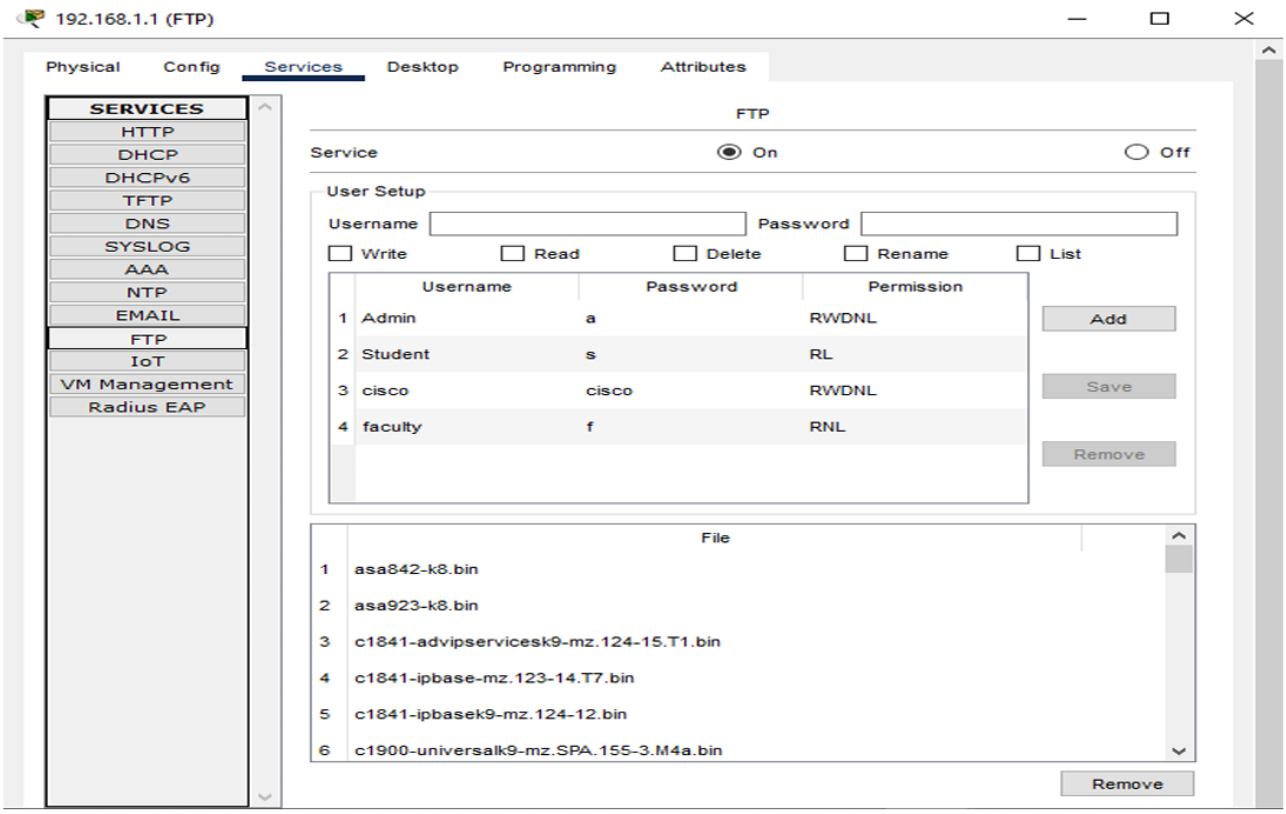


3.1 Circuit Design of FTP

IP Address:

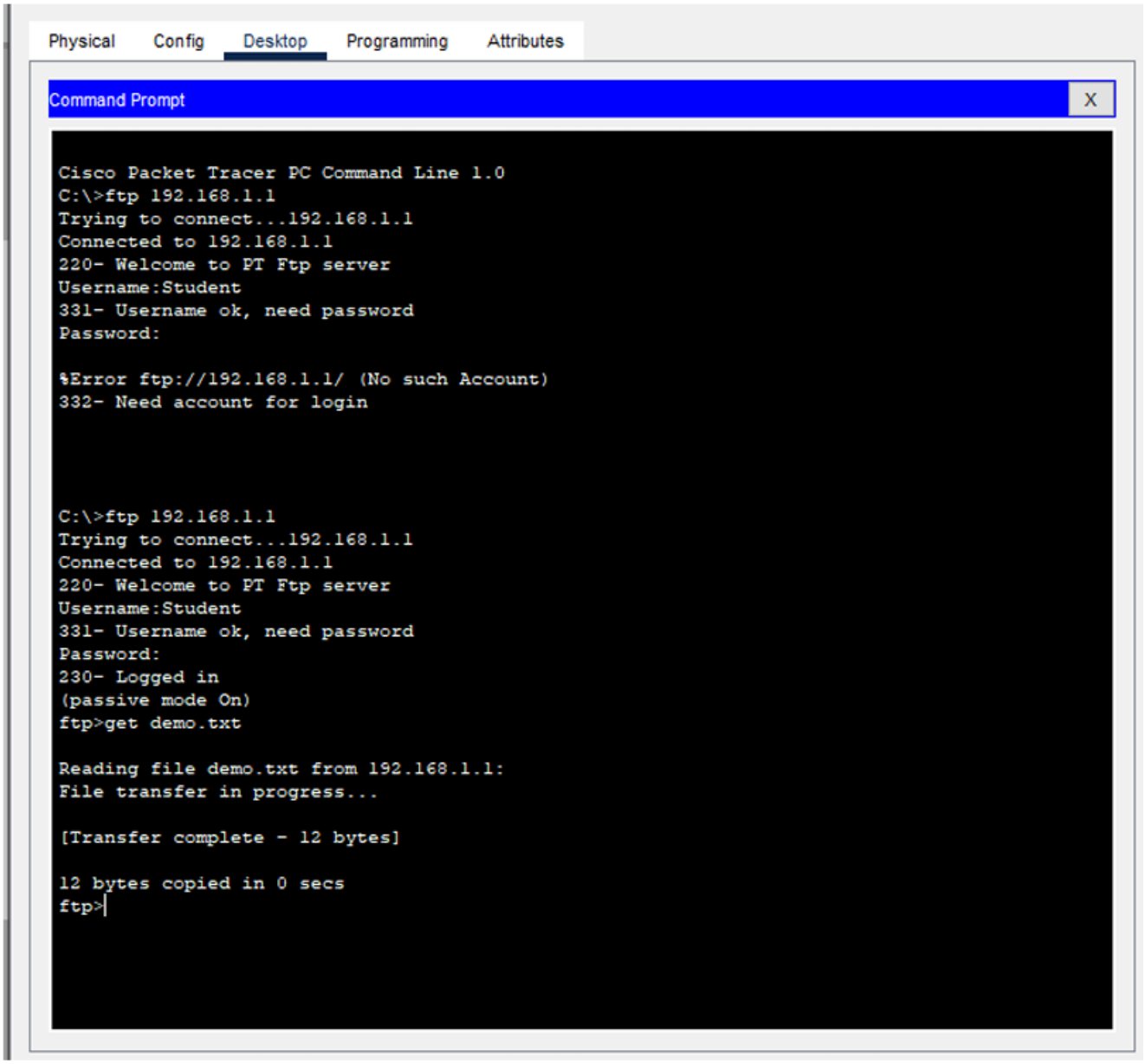
Device	IP Address	Subnet Mask	Default Gateway
FTP Server	192.168.1.1	255.255.255.0	N/A
PC0 (Admin)	192.168.1.2	255.255.255.0	N/A
PC1 (Student)	192.168.1.3	255.255.255.0	N/A
PC2 (Faculty)	192.168.1.4	255.255.255.0	N/A

Date: 16/07/2026



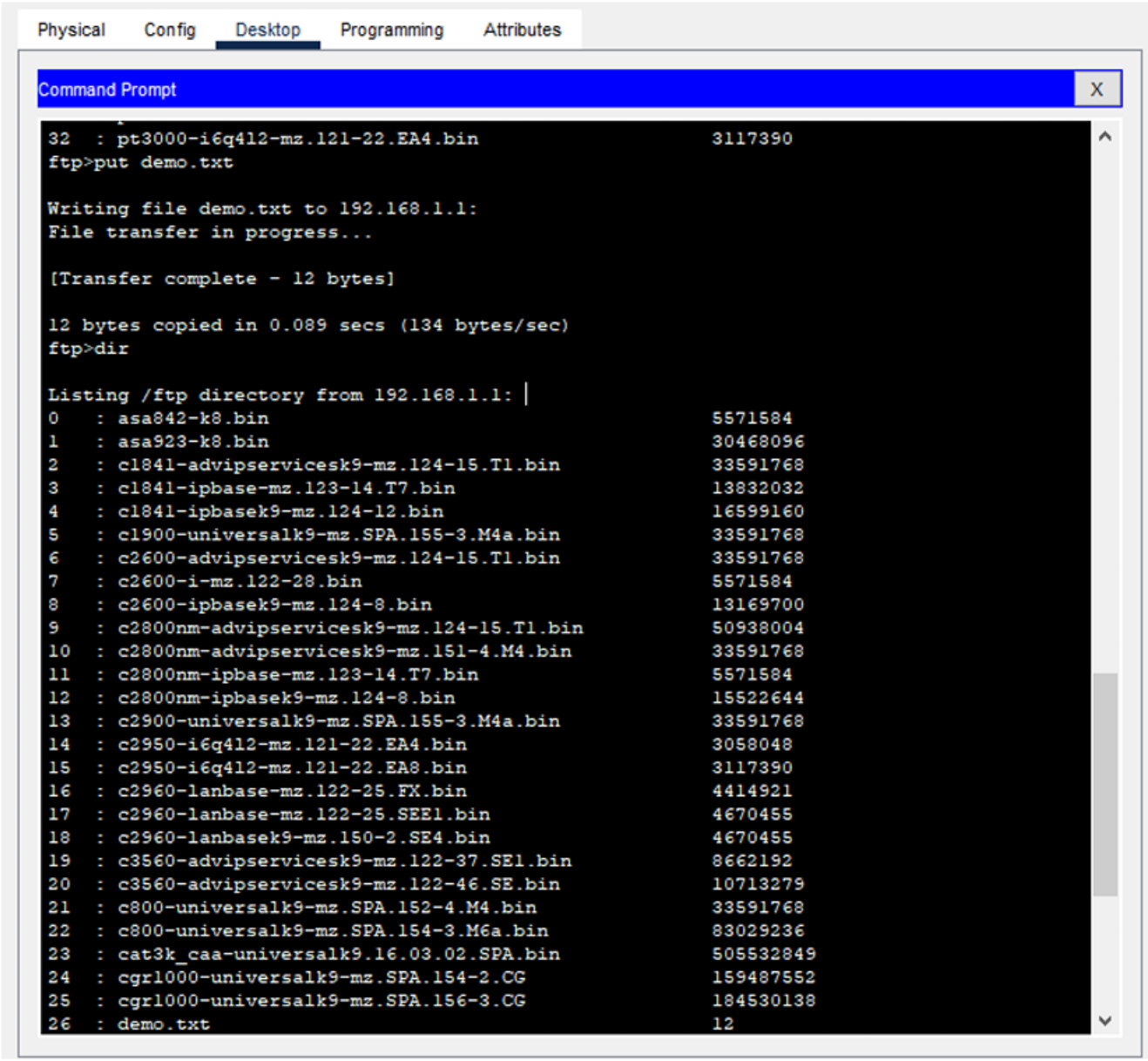
3.2 FTP Server Service

Date: 16/07/2026



3.3 Login User in FTP Server

Date: 16/07/2026



3.4 Put and view Directory

Date: 16/07/2026

➔ **Uses OF this Server in Real world examples:**

1 DNS Server

Real-Life Uses

- Accessing websites using names instead of IP addresses.
- Email servers use DNS to deliver emails.
- Online gaming servers use DNS for easy connections.
- Cloud services and mobile applications rely on DNS.

Example

When you type www.youtube.com into your browser:

1 Your computer sends a DNS request.

2 The DNS server finds YouTube's IP address.

3 Your browser connects to that IP address and loads the website.

2 DHCP Server

Real-Life Uses

- Home Wi-Fi routers assign IP addresses automatically.
- Schools and colleges manage hundreds of computers.
- Offices automatically configure employee computers.
- Hotels, airports, and cafés provide automatic Internet access.

Example

When you connect your smartphone to your home Wi-Fi:

1. The router (DHCP Server) detects the new device.
2. It automatically assigns an IP address (for example, 192.168.1.25).
3. You can immediately access the Internet without manual configuration.

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3 FTP Server

Real-Life Uses

- Uploading websites to a web hosting server.
- Sharing project files in companies.
- Downloading software from FTP servers.
- Backing up files to a remote server.

Example

- A college stores study materials on an FTP server.
- Students log in using a username and password.
- They download notes, assignments, and lab manuals.